Non-isothermal unipolar simulation of an $n^+ - n - n^+$ device

Matteo Zorat

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0 Note to the reader

To ensure a stable and consistent starting point, the non-isothermal solution is initialized from a preliminary isothermal simulation computed with a strict convergence tolerance of 10^{-10}.
Unless stated otherwise, the numerical framework utilizes the following reference parameters: a uniform mesh of $N = 1000$ *nodes, an energy relaxation time of* $\tau_E = 0.2$ *ps, a device length of* $L = 3 \times 10^{-6}$ *m, a contact region length of* $L_c = 0.45 \times 10^{-6}$ *m, and a transition smoothing distance of* $\sigma = 30 \times 10^{-9}$ *m.*
For the simulated bias range and grid parameters evaluated in this project, the numerical framework proved stable, and no under-relaxation was required ($\omega = 1$*).*

1 Equations and Boundary Conditions

$$\begin{cases} -\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\varepsilon\,\dfrac{\mathrm{d}\varphi}{\mathrm{d}x}\right) = q\big(N_D - n\big), & \text{(1a)} \\ \dfrac{\mathrm{d}J_n}{\mathrm{d}x} = 0, & \text{(1b)} \\ J_n = q\mu_n\left(-n\,\dfrac{\mathrm{d}\varphi}{\mathrm{d}x} + \dfrac{\mathrm{d}(n\vartheta)}{\mathrm{d}x}\right), & \text{(1c)} \\ \dfrac{\mathrm{d}S_n}{\mathrm{d}x} = J_n E - \dfrac{3}{2}\,\dfrac{qn(\vartheta - V_T)}{\tau_E}, & \text{(1d)} \\ S_n = -\lambda_n\,\dfrac{\mathrm{d}\vartheta}{\mathrm{d}x} - \dfrac{5}{2}\,\vartheta\,J_n, & \text{(1e)} \end{cases}$$

where $E = -\mathrm{d}\varphi/\mathrm{d}x$ is the electric field, S_n is the electron energy flux, τ_E is the energy relaxation time, and λ_n is the thermal conductivity coefficient in the ϑ-variable. In the basic model, the Wiedemann–Franz-type closure is adopted:

$$\lambda_n = c_\lambda\, q\mu_n n\,\vartheta, \qquad c_\lambda = \tfrac{5}{2}.$$

Equation (1a) is Poisson's equation, coupling the potential to the net charge density $q(N_D - n)$. Equation (1b) expresses charge conservation. Equation (1c) is the generalized drift-diffusion current, which reduces to the standard isothermal form when $\vartheta = V_T$. Equations (1d)–(1e) form the electron energy balance: the divergence of the energy flux equals the Joule source $J_n E$ minus the energy lost to the lattice through collisions at rate $3qn(\vartheta - V_T)/(2\tau_E)$.

1.1 Boundary conditions

Ohmic, thermally equilibrated contacts are imposed at both ends:

$$\begin{aligned} &\varphi(0) = 0, && \varphi(L) = V_a, \\ &n(0) = N_D^+, && n(L) = N_D^+, \\ &\vartheta(0) = V_T, && \vartheta(L) = V_T. \end{aligned} \tag{2}$$

The electron temperature is pinned to the lattice temperature at the contacts ($T_n = T_L$ at $x = 0$ and $x = L$).

1.2 Doping profile

The smoothed doping profile is

$$N_D(x) = N_D^{\text{low}} + \left(N_D^{+} - N_D^{\text{low}}\right)\left[S_L(x) + S_R(x)\right], \tag{3}$$

with

$$S_L(x) = \tfrac{1}{2}\left[1 - \tanh\left(\frac{x - L_c}{\sigma}\right)\right], \qquad S_R(x) = \tfrac{1}{2}\left[1 + \tanh\left(\frac{x - (L - L_c)}{\sigma}\right)\right],$$

where $N_D^{\text{low}} = 10^{22}\,\text{m}^{-3}$, $N_D^{+} = 10^{24}\,\text{m}^{-3}$, $L_c = 0.45\,\mu\text{m}$, and $\sigma = 30\,\text{nm}$.

2 Discrete Scharfetter–Gummel Fluxes

To ensure numerical stability, enforce local conservation laws, and maintain strict positivity safeguards for the carrier concentration and electron temperature on grids, the continuous fluxes are discretized using Scharfetter–Gummel.

2.1 Discrete Current Density Flux

By treating the current density J_n as a constant value across each individual mesh edge $[x_i, x_{i+1}]$ and utilizing an averaging of the thermal voltage variable, the discrete Scharfetter–Gummel current expression on the edge $i + 1/2$ is formulated as:

$$J_{n,i+1/2} = \frac{q\mu_n}{h_i}\vartheta_{i+1/2}\left[B(\Delta_{i+1/2})n_{i+1} - B(-\Delta_{i+1/2})n_i\right] \tag{4}$$

where the normalized, localized potential drop $\Delta_{i+1/2}$ across the mesh element is expressed as:

$$\Delta_{i+1/2} = \frac{\phi_{i+1} - \phi_i}{\vartheta_{i+1/2}}$$

To properly capture thermal boundaries, the localized thermal voltage $\vartheta_{i+1/2}$ on the edge is evaluated using the logarithmic mean approximation:

$$\vartheta_{i+1/2} = L(\vartheta_i, \vartheta_{i+1}) = \frac{\vartheta_{i+1} - \vartheta_i}{\ln(\vartheta_{i+1}) - \ln(\vartheta_i)}$$

When $\vartheta_i \approx \vartheta_{i+1}$, the logarithmic mean simplifies to the standard arithmetic average using a local Taylor expansion step to retain precision.

2.2 Discrete Energy Flux

Substituting the closure model for λ_n directly into the continuous energy flux expression yields a structure that can be stabilized into the drift-diffusion current equation. The resulting discrete electron energy flux $S_{n,i+1/2}$ assigned to the mesh edge is expressed as:

$$S_{n,i+1/2} = -c_\lambda \frac{q\mu_n}{h_i}\vartheta_{i+1/2}^2\left[B(\Gamma_{i+1/2})\frac{n_{i+1}}{\vartheta_{i+1}} - B(-\Gamma_{i+1/2})\frac{n_i}{\vartheta_i}\right] - \frac{5}{2}\vartheta_{i+1/2}J_{n,i+1/2} \tag{5}$$

where $\Gamma_{i+1/2}$ handles the local advective transport weighting factor dictated by the coupled interaction of the electric field and the localized current flow across the grid interval.

3 Gummel Iteration Scheme

The non-linear coupling between fields, carrier profiles, and temperatures is resolved via a decoupled Gummel iteration map:

Step 1: Linearized Poisson Equation

The electrostatic potential is updated from φ^k to φ^{k+1} by solving a Gummel-linearized version of the Poisson equation. The carrier concentration is modeled with a standard Boltzmann-like state mapping to enhance numerical stability:

$$-\frac{d}{dx}\left(\epsilon\frac{d\varphi^{k+1}}{dx}\right) + qn^k\frac{\varphi^{k+1} - \varphi^k}{\vartheta^k} = q(N_D - n^k) \tag{6}$$

This formulation yields a linear tridiagonal system after finite-difference discretization, providing robust convergence properties even under strong space-charge conditions.

Step 2: Electron Continuity Equation

Using the newly computed potential profile φ^{k+1} along with the previous iteration's thermal voltage profile ϑ^k, the electron continuity equation is solved to yield the updated carrier distribution n^{k+1}:

$$\frac{d}{dx}\left(J_n(\varphi^{k+1}, n^{k+1}, \vartheta^k)\right) = 0 \tag{7}$$

This system is linear with respect to the nodal electron densities n_i^{k+1} when using the discrete Scharfetter–Gummel flux setup.

Step 3: Energy Balance Equation

Next, the electron thermal voltage profile is updated to ϑ^{k+1} by solving the energy balance equation using the updated potential φ^{k+1} and the newly updated electron density distribution n^{k+1}:

$$\frac{d}{dx}\left(S_n(\varphi^{k+1}, n^{k+1}, \vartheta^{k+1})\right) = J_n^{k+1}E^{k+1} - \frac{3}{2}qn^{k+1}\frac{\vartheta^{k+1} - V_T}{\tau_E} \tag{8}$$

The discrete flux and the linear relaxation loss term form a linear system for the unknown thermal voltages ϑ_i^{k+1} at the internal nodes.

Step 4: Damping

To guarantee convergence under high bias conditions, parameters $\omega_\varphi, \omega_n, \omega_T \in [0.2, 1.0]$ can be dynamically applied to damp the updates before proceeding to the next iteration loop:

$$\begin{aligned}
\varphi^{k+1} &\leftarrow (1-\omega_\varphi)\varphi^k + \omega_\varphi\varphi^{k+1} \\
n^{k+1} &\leftarrow (1-\omega_n)n^k + \omega_n n^{k+1} \\
\vartheta^{k+1} &\leftarrow (1-\omega_T)\vartheta^k + \omega_T\vartheta^{k+1}
\end{aligned}$$

Step 5: Convergence

The iteration loop terminates when the maximum relative infinity norm of the updates across all three primary variables falls below the specified tolerance:

$$\max\left\{\frac{\|\varphi^{k+1} - \varphi^k\|_\infty}{\max(V_T, \|\varphi^{k+1}\|_\infty)}, \frac{\|n^{k+1} - n^k\|_\infty}{\|n^{k+1}\|_\infty}, \frac{\|\vartheta^{k+1} - \vartheta^k\|_\infty}{\|\vartheta^{k+1}\|_\infty}\right\} < 10^{-8} \tag{9}$$

4 Thermal equilibrium

We use the thermal equilibrium condition ($V_a = 0$) as the primary benchmark for physical and numerical consistency. Under these conditions, the system must remain in a state of minimum entropy.

Regarding the total electronic current density (J_n), the net value must equal zero. Physically, this represents the exact spatial cancellation of the drift component by the diffusion component. The absence of an external bias ensures that no net charge transport occurs across the device.

The carrier temperature (T_n) should show zero deviation from the lattice temperature (T_L).

Finally, the electron density (n) and the electrostatic potential (φ) exhibit complex dynamics in the presence of spatial variations. While uniformly doped bulk regions maintain local charge neutrality ($n = N_D^+$), a localized space-charge region is physically expected near junctions.

4.1 Numerical results

By running our Matlab simulation we obtain the following results:

Current density

$$\overline{J_n} = -1.795862 \cdot 10^{-5}\frac{\mathrm{A}}{\mathrm{m}^2} \qquad \mathrm{spread}(J_n) = 6.514673 \cdot 10^{-4}\frac{\mathrm{A}}{\mathrm{m}^2}$$

The calculated current density is close to zero, yet its absolute value remains well above machine epsilon due to the numerical formulation. The current is derived using the Scharfetter-Gummel discretization, which balances the opposing drift and diffusion components at equilibrium. While this cancellation is exact in a continuous system, it is limited on a discrete grid by the spatial truncation error of the mesh. Because the grid has a finite density, evaluating these spatial derivatives inevitably yields a tiny, non-zero numerical residual.

Temperature

$$\overline{T_n} = 300.0000 \text{ K} \qquad \max_x \left(\frac{T_n - T_L}{T_L} \right) = 6.063298 \cdot 10^{-15} \approx \text{eps}$$

As expected, the electron temperature remains completely uniform and perfectly thermalized at 300 K throughout the device, rendering the maximum relative deviation computationally zero. In the "Temperature" subplot, this behavior is illustrated by the perfectly horizontal line where the non-isothermal and isothermal curves completely overlap without a single node of departure.

Electron concentration

$$\operatorname*{mean}_x \left(\frac{n}{N_D(x)} \right) = 1.029745 \qquad \max_x \left(\frac{n}{N_D(x)} \right) = 1.550598$$

The global charge neutrality is preserved on average as the mean electron concentration closely tracks the background doping. However, a localized discrepancy arises within the graded transition zones ($x \approx 0.45$ μm and $x \approx 2.55$ μm), where the maximum $n/N_D(x)$ ratio reaches 1.55. This behavior is captured in the "Quasi-neutrality ratio" and "Electron density" subplots, which show two symmetric, peaks exactly where the doping profile varies most rapidly.

Voltage potential

$$\operatorname*{mean}_x \left(\varphi - V_T \log \frac{N_D(x)}{N_D^+} \right) = 6.561661 \cdot 10^{-4} \text{ V}$$

$$\max_x \left(\varphi - V_T \log \frac{N_D(x)}{N_D^+} \right) = 1.133974 \cdot 10^{-2} \text{ V}$$

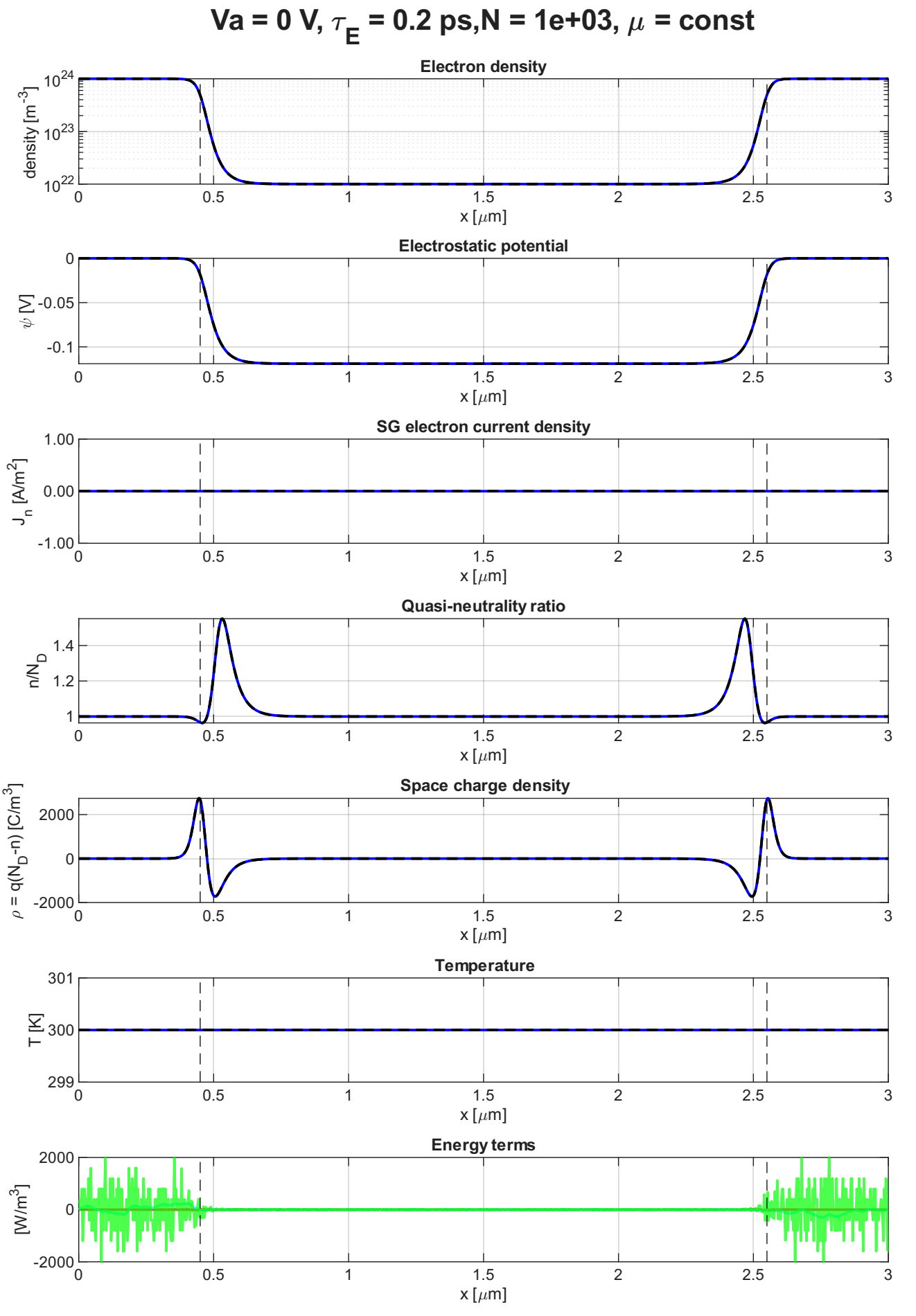
$$\frac{\max_x \left| \varphi - V_T \log \frac{N_D(x)}{N_D^+} \right|}{\max_x \left| V_T \log \frac{N_D(x)}{N_D^+} \right|} = 9.524960 \cdot 10^{-2}$$

Regarding the electrostatic potential, evaluating the simulation against a local charge neutrality baseline yields a maximum localized deviation of approximately 11.34 mV, which corresponds to a minor 9.52% relative maximum error. This low deviation captures the residual physical smoothing of the potential across the graded interface, visible in the "Electrostatic potential" subplot.

Furthermore, the spatial mean of this deviation leaves a low residual of only 0.66 mV. This tiny non-zero average is a direct consequence of the slight spatial asymmetry in the local Debye lengths across the graded junction. While the heavily doped contact sides exhibit a narrower screening width, the lightly doped channel allows the spilled-over electrons to form a slightly more extended potential tail.

Energy balance

As just seen, the total current density is null (almost) and the electron temperature completely thermalizes to the lattice baseline across the entire device domain. Analytically, this eliminates both Joule heating and collision losses, meaning the spatial derivative of the electron energy flux (dS_n/dx) must equal zero everywhere. Consequently, the high-frequency oscillations between -2000 and 2000 W/m^3 visible near the device boundaries in the "Energy terms" subplot are a purely numerical artifact. Because the electron concentration peaks at 10^{24} m^{-3} within highly doped contact zones, microscopic truncation errors stemming from the discrete grid discretization close to machine precision are amplified by twenty-four orders of magnitude during finite-difference evaluation.


Va = 0 V, τ_E = 0.2 ps,N = 1e+03, μ = const
Electron density
density [m^{-3}]
10^{24}
10^{23}
10^{22}
0
0.5
1
1.5
2
2.5
3
x [μm]
Electrostatic potential
ψ [V]
0
-0.05
-0.1
x [μm]
SG electron current density
J$_n$ [A/m^2]
1.00
0.00
-1.00
x [μm]
Quasi-neutrality ratio
n/N$_D$
1.4
1.2
1
x [μm]
Space charge density
ρ = q(N$_D$-n) [C/m^3]
2000
0
-2000
x [μm]
Temperature
T [K]
301
300
299
x [μm]
Energy terms
[W/m^3]
2000
0
-2000
x [μm]
Non-isothermal
Isothermal
$J_n \cdot E$
$\frac{3}{2}qn\frac{\theta - V_T}{\tau_E}$
$\frac{dS_n}{dx}$

5 Voltage sweep comparison

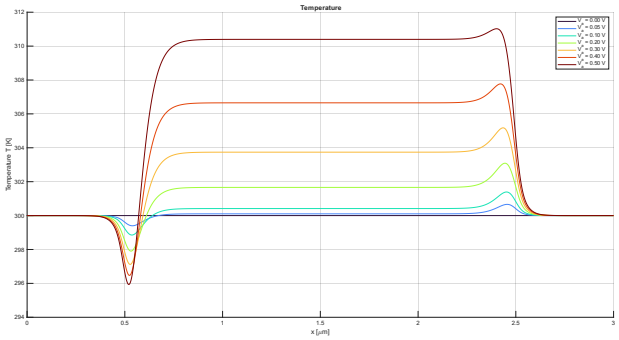


Figure 1: Temperature as function of bias

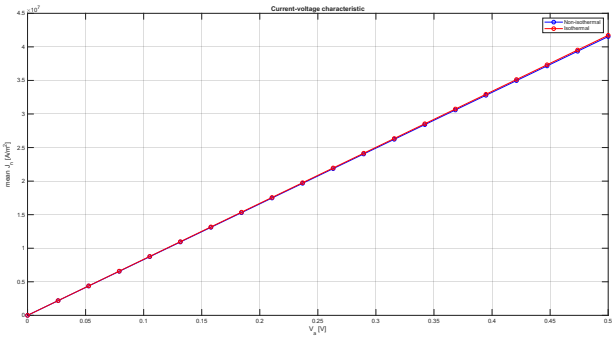


Figure 2: I-V curve

During the voltage sweep from 0.050 V to 0.500 V, spatial profiles reveal a highly asymmetric thermal distribution. This behavior is initiated by a localized cooling effect below the 300 K lattice temperature near the injecting $n^+ - n$ junction. In this region, electrons expand into the lower-doped channel against a positive built-in electric field. Because the conventional electron current flows to the right, the resulting Joule heating term becomes negative ($J_n \cdot E < 0$), acting as a local heat sink that extracts kinetic energy from the carrier. In contrast, a thermal peak develops near the extracting $n - n^+$ junction, where the electric field is strongly negative. When multiplied by the negative J_n, it yields a positive Joule heating term ($J_n \cdot E > 0$), creating a heat source that heats the carriers before they thermalize back to the 300 K boundary equilibrium condition upon entering the highly doped contact.

V_a [V]	$\frac{\max(\lvert u\rvert)}{v_{\text{sat}}}$	$\frac{\max(\mu\lvert\nabla\varphi_F\rvert)}{v_{\text{sat}}}$	$\max(T_n)$	$\frac{\text{spread}(J_n)}{J_{\text{scale}}}$	$\frac{\text{spread}(J_n)}{\lvert J_{\text{mean}}\rvert}$
0.000	$6.193 \cdot 10^{-14}$	$6.008 \cdot 10^{-14}$	300.0	$4.827 \cdot 10^{-12}$	$4.359 \cdot 10^{1}$
0.050	$2.592 \cdot 10^{-2}$	$2.592 \cdot 10^{-2}$	300.7	$2.786 \cdot 10^{-9}$	$9.264 \cdot 10^{-8}$
0.100	$5.183 \cdot 10^{-2}$	$5.183 \cdot 10^{-2}$	301.4	$4.507 \cdot 10^{-9}$	$7.493 \cdot 10^{-8}$
0.200	$1.037 \cdot 10^{-1}$	$1.037 \cdot 10^{-1}$	303.1	$6.438 \cdot 10^{-9}$	$5.351 \cdot 10^{-8}$
0.300	$1.555 \cdot 10^{-1}$	$1.555 \cdot 10^{-1}$	305.2	$1.557 \cdot 10^{-8}$	$8.626 \cdot 10^{-8}$
0.400	$2.074 \cdot 10^{-1}$	$2.074 \cdot 10^{-1}$	307.8	$1.710 \cdot 10^{-8}$	$7.107 \cdot 10^{-8}$
0.500	$2.592 \cdot 10^{-1}$	$2.592 \cdot 10^{-1}$	311.0	$2.868 \cdot 10^{-8}$	$9.534 \cdot 10^{-8}$

Table 1: Voltage sweep data

The validity of the constant low-field mobility assumption is monitored via the edge velocity indicator $\max(|u_n|)/v_{\text{sat}}$, which remains comfortably below the 0.2 threshold up to 0.300 V, but reaches a transition range between 0.2 and 0.3 at higher voltages, indicating that the constant-mobility model is reaching its limits. Despite this electron heating, the current-voltage characteristic shows a nearly perfect overlap between the isothermal and the non-isothermal model, meaning macroscopic transport is dominated by the drift-diffusion mechanisms.

5.1 Proof of model validity

We can write the effect of velocity saturation on mobility as:

$$\mu_n(E) = \frac{\mu_0}{1 + \dfrac{\mu_0 |E|}{v_{sat}}} = \frac{\mu_0}{1 + x}$$

The electron velocity results:

$$u = \mu(E)E = \frac{\mu_0 E}{1 + x}$$

Calling the velocity in the case of constant mobility $u_0 = \mu_0 E$, then we can compute the absolute error

$$\varepsilon = \frac{u_0 - u}{u_0} = 1 - \frac{1}{1 + x} = \frac{x}{1 + x} \qquad \text{with} \quad x = \frac{\mu_0 |E|}{v_{sat}} = \frac{|u_n|}{v_{sat}}$$

By computing the error for $\max(x) = 0.3$ we have $\varepsilon \approx 0.2307$, and so we are overestimating our mobility by a factor around 23%
In our case we have a maximum equal to 0.2592 for $V_A = 0.5V$ which leads to an error of 20.6%. Which is acceptable because this velocity overestimation is strictly limited to the narrow zone near the $n - n^+$ junction where both the electric field and the electron temperature peak. Outside this high-field, high-temperature region, the total current is regulated by the injection barrier and the bulk channel, where fields are low and the constant-mobility assumption remains highly accurate. A visualization of the phenomenon is provided in the following section.

5.2 Mobility dependence

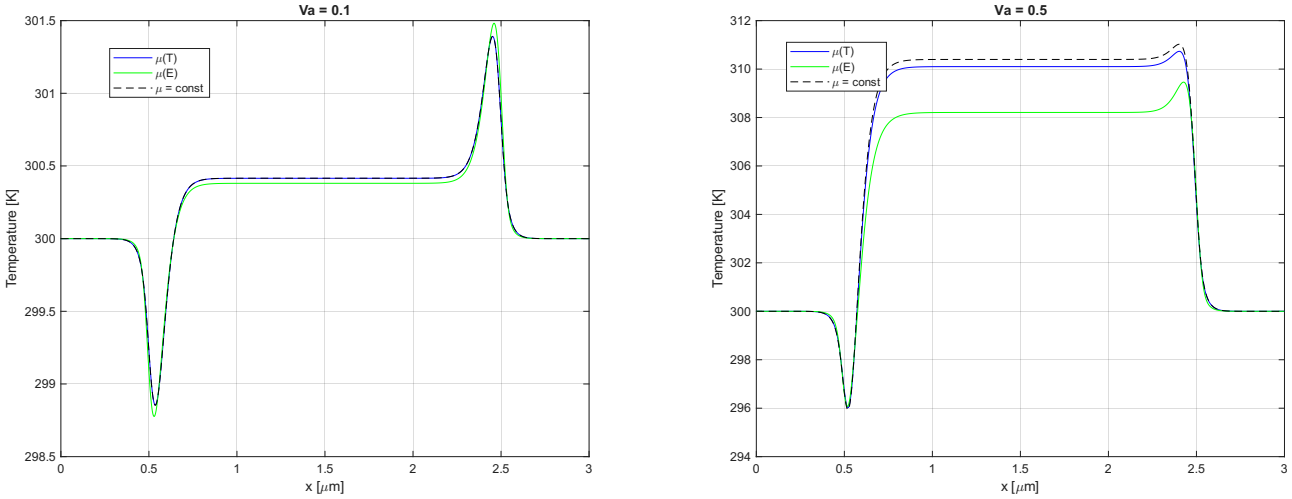


Figure 3: Temperature

To transition from the simplified baseline toward a accurate representation of the device, the constant low-field mobility assumption must be replaced by formulations that account for temperature and electric field dependencies.

Incorporating a temperature-dependent mobility model via the law $\mu_n(T_n) = \mu_0(T_L/T_n)^\gamma$ introduces the effect of enhanced thermal carrier scattering at elevated electron energies. However, looking at the simulation data, electrons experience a minor peak temperature of 311.0 K at the maximum bias of $V_a = 0.500$ V, which represents a thermal increase of only 11.0 K above the lattice equilibrium ($T_L = 300$ K). This means the pure thermal degradation factor remains close to unity, reducing local mobility by less than 7% at the hottest point of the channel.

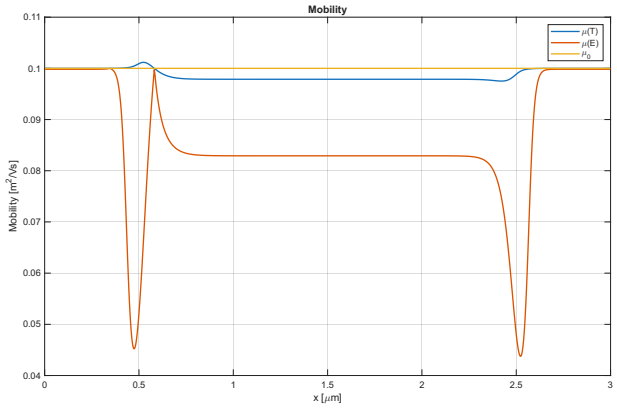


Figure 4: Mobility for $V_A = 0.4V$

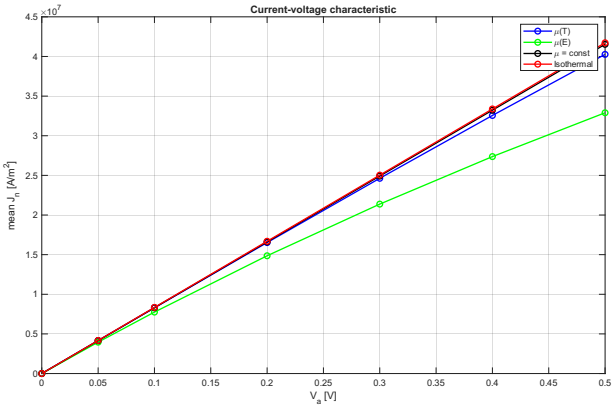
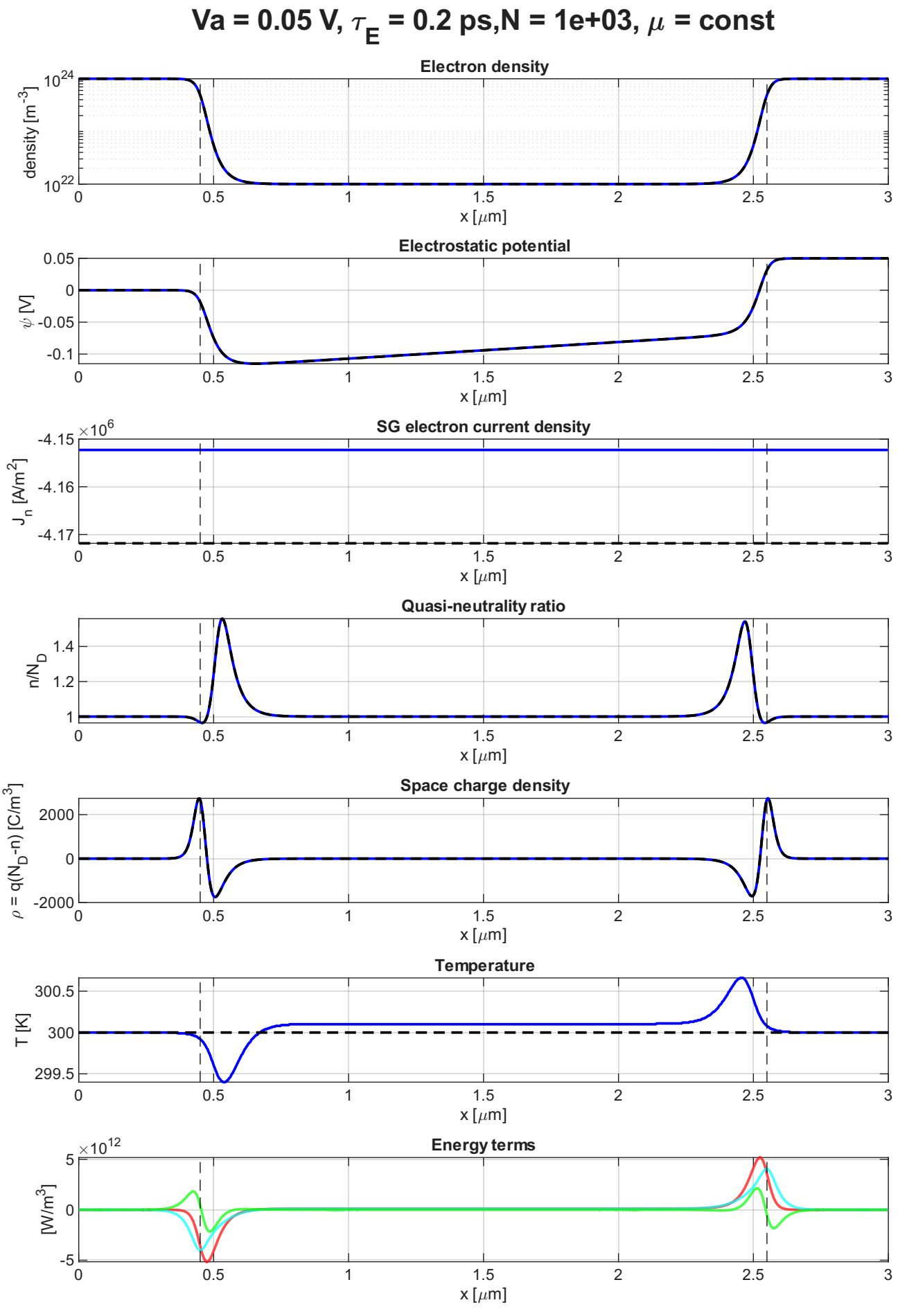
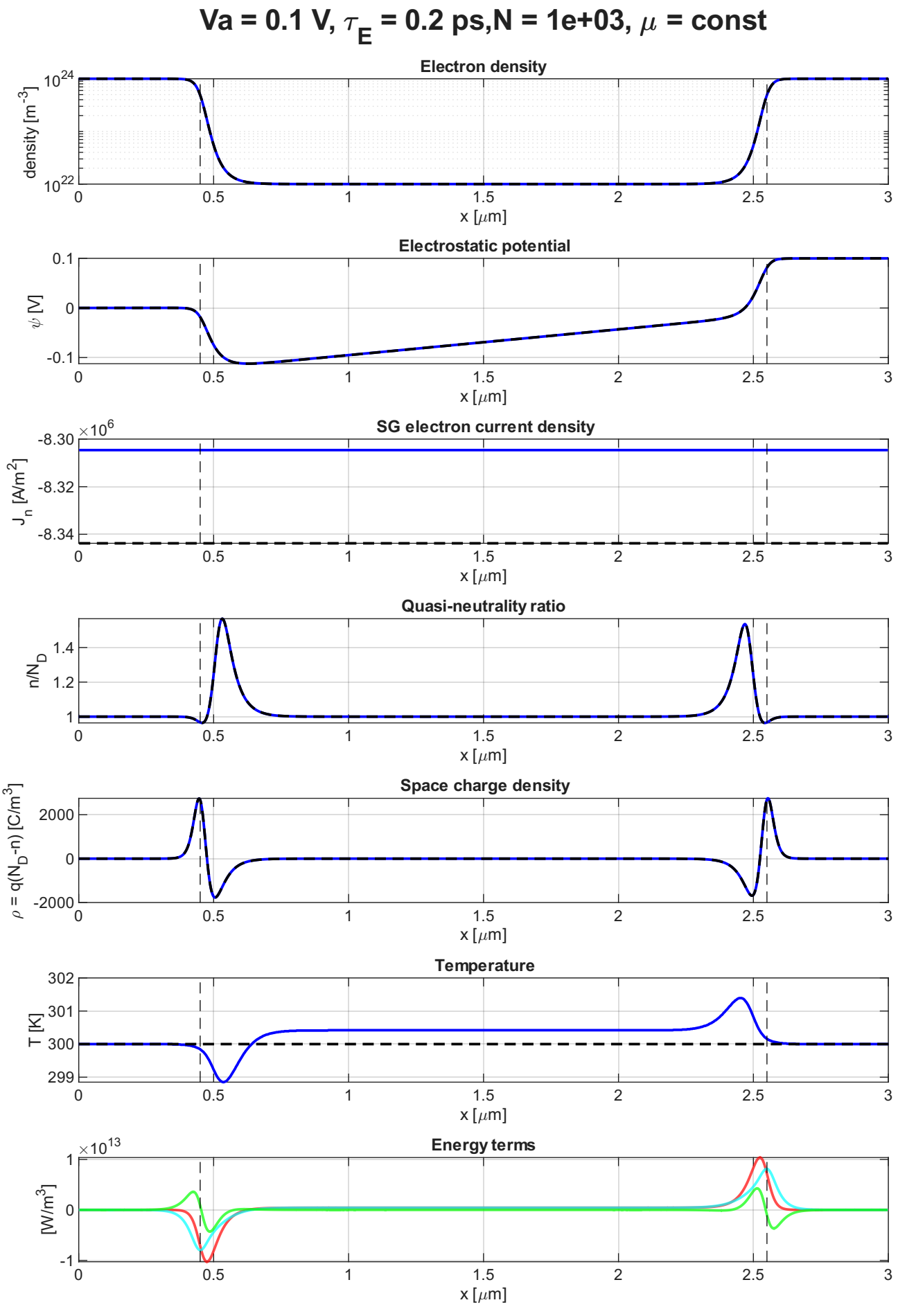


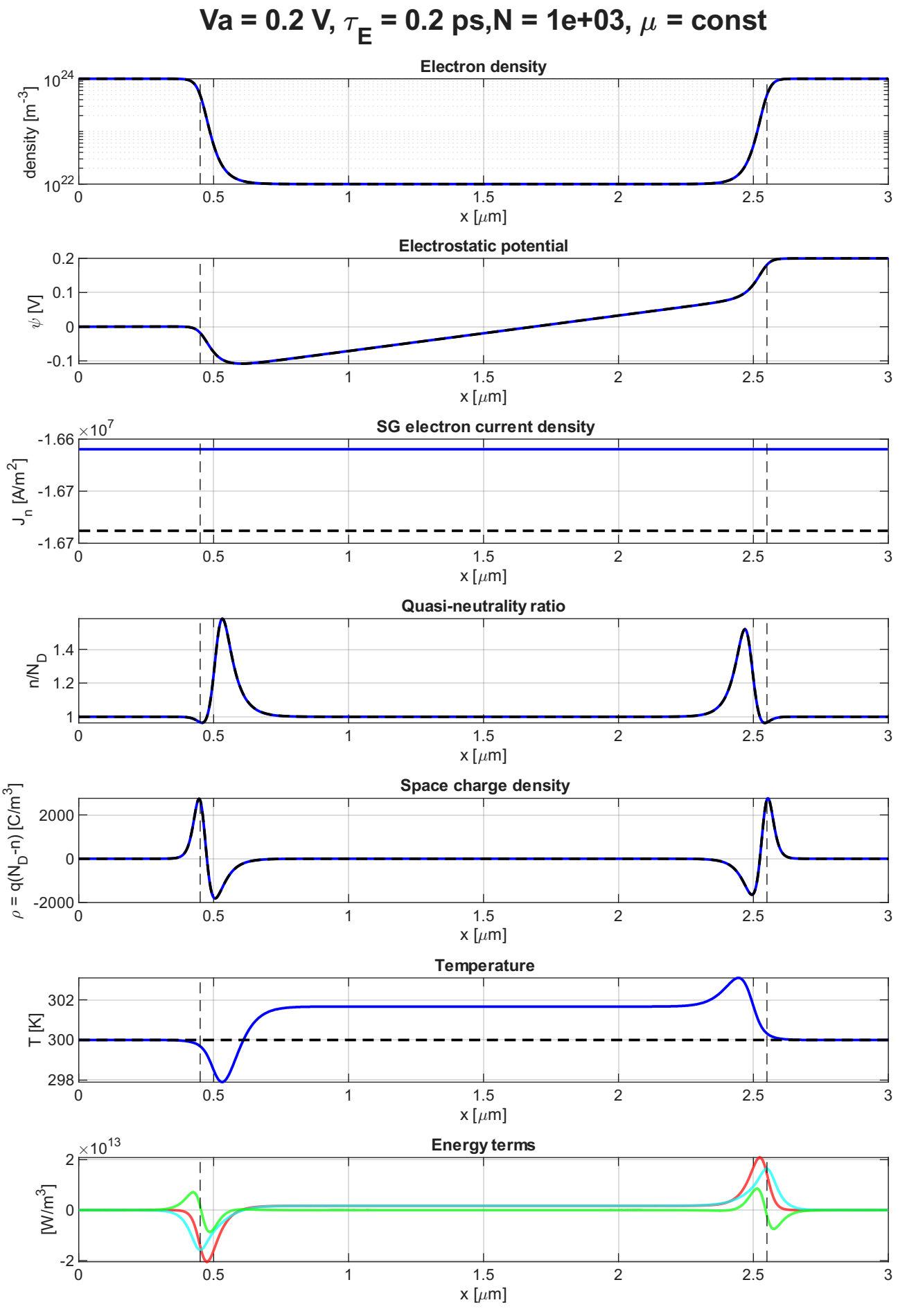
Figure 5: I-V curve

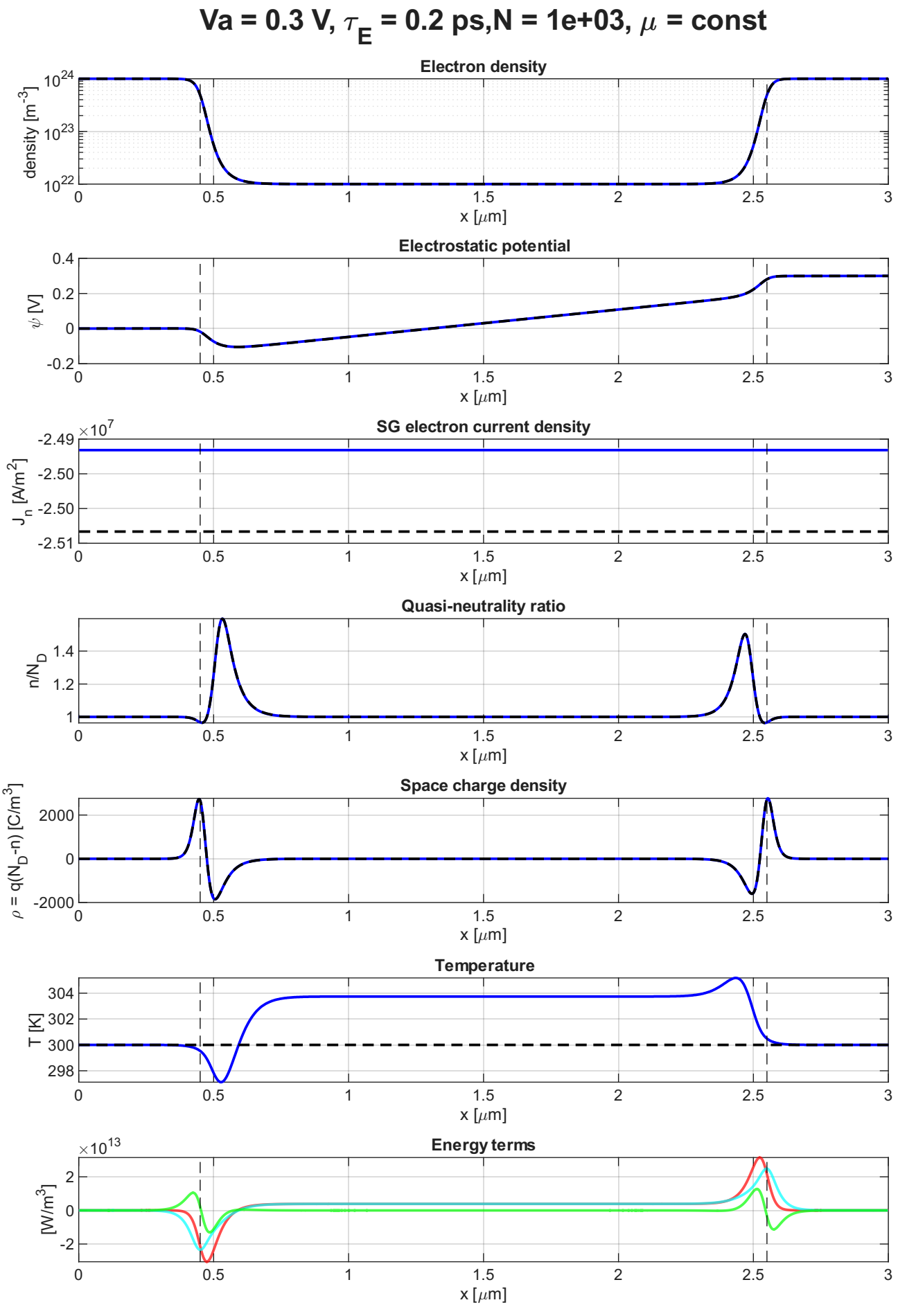
On the other hand, implementing the field-dependent mobility correction $\mu_n(E) = \mu_0/(1 + \mu_0|E|/v_{\text{sat}})$ introduces a dominant physical correction. Because the intense localized potential drop near the extracting junction pushes the edge velocity indicator $\max(|u_n|)/v_{\text{sat}}$ to 0.2592, the model overestimates the true carrier velocity by 20.6%.

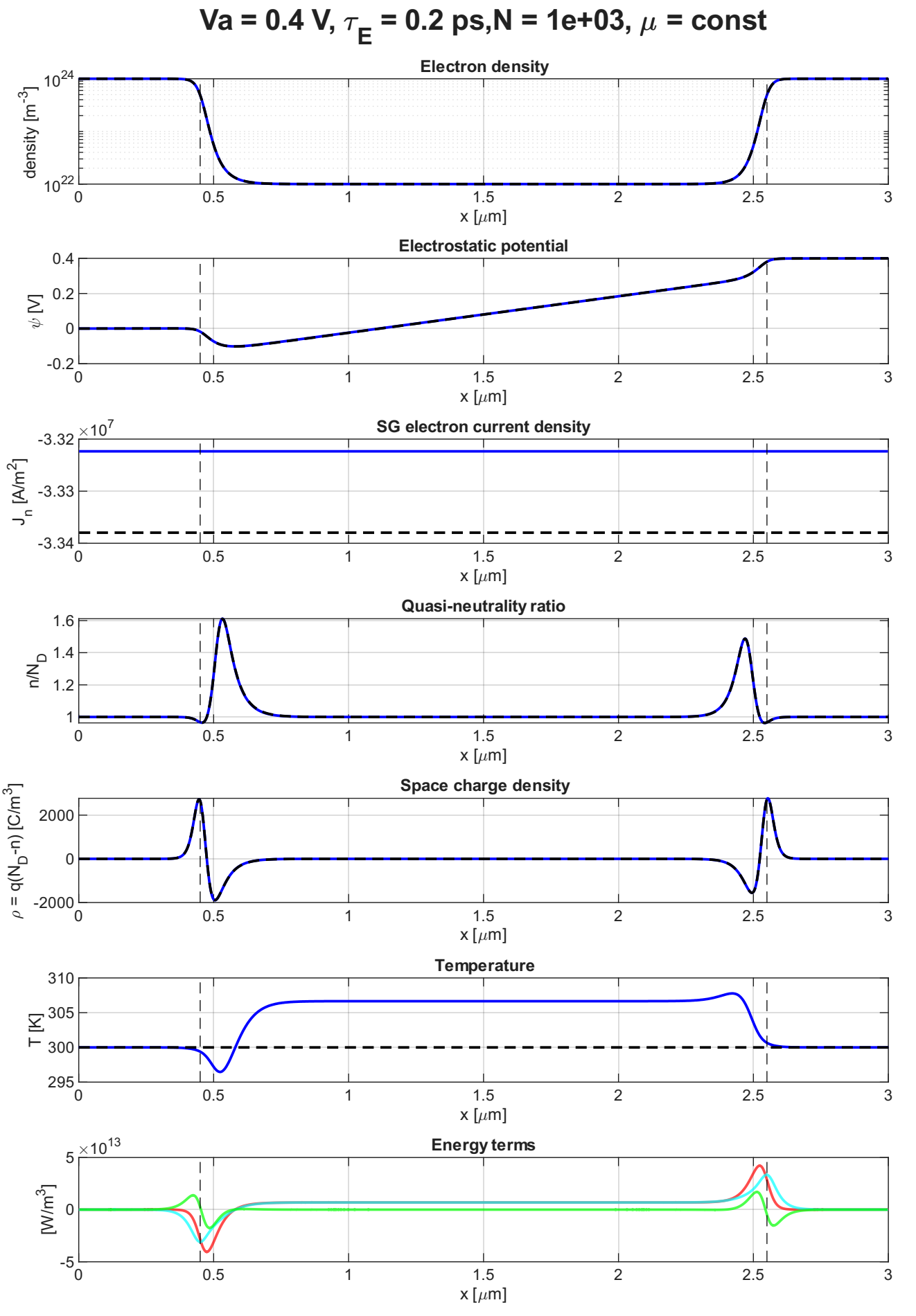
Ultimately, relying only on a field-dependent mobility isolates velocity saturation from the actual energy relaxation process, failing to capture how hot electrons actively suppress current via thermal scattering. However, one cannot simply combine the temperature-dependent and field-dependent models to fix this. Both are phenomenological approximations describing the exact same microscopic mechanism (scattering effect). Combining them would improperly double-count these scattering events. Therefore, to accurately simulate submicron devices subjected to both high fields and severe electron heating, these simplified macroscopic approximations must be abandoned. Instead, it is necessary to resort to rigorous full hydrodynamic models or solve the Boltzmann Transport Equation directly using Monte Carlo particle simulations.

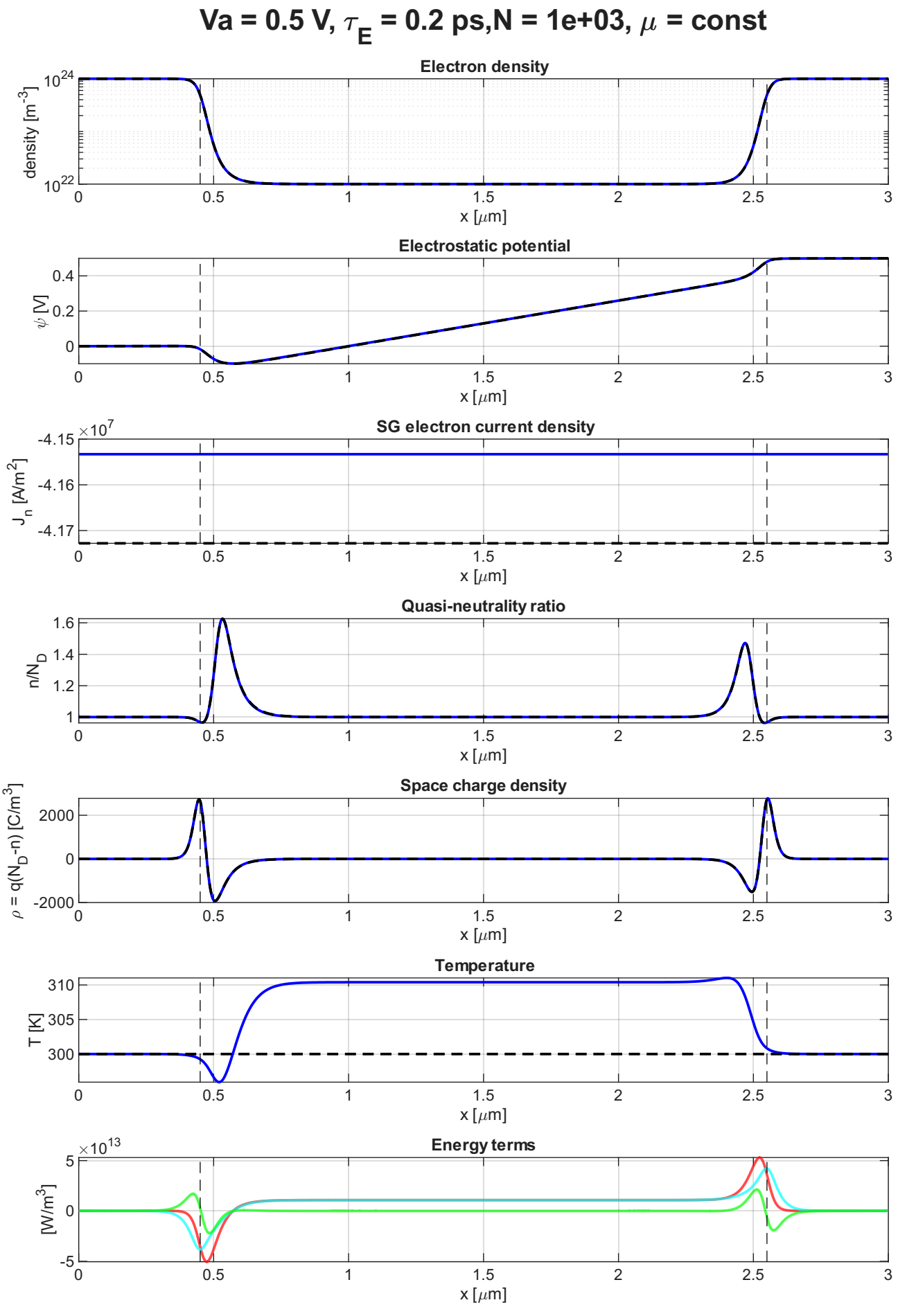

Va = 0.05 V, τ_E = 0.2 ps,N = 1e+03, μ = const
Electron density
density [m^{-3}]
10^{24}
10^{22}
x [μm]
Electrostatic potential
ψ [V]
0.05
0
-0.05
-0.1
x [μm]
SG electron current density
$\times 10^6$
J_n [A/m^2]
-4.15
-4.16
-4.17
x [μm]
Quasi-neutrality ratio
n/N$_D$
1.4
1.2
1
x [μm]
Space charge density
ρ = q(N$_D$-n) [C/m^3]
2000
0
-2000
x [μm]
Temperature
T [K]
300.5
300
299.5
x [μm]
Energy terms
$\times 10^{12}$
[W/m^3]
5
0
-5
x [μm]
0
0.5
1
1.5
2
2.5
3
Non-isothermal
Isothermal
$J_n \cdot E$
$\frac{3}{2}qn\frac{\theta - V_T}{\tau_E}$
$\frac{dS_n}{dx}$


Va = 0.1 V, τ_E = 0.2 ps,N = 1e+03, μ = const
Electron density
density [m^{-3}]
10^{24}
10^{22}
x [μm]
Electrostatic potential
ψ [V]
0.1
0
-0.1
x [μm]
SG electron current density
$\times 10^6$
J_n [A/m^2]
-8.30
-8.32
-8.34
x [μm]
Quasi-neutrality ratio
n/N$_D$
1.4
1.2
1
x [μm]
Space charge density
ρ = q(N$_D$-n) [C/m^3]
2000
0
-2000
x [μm]
Temperature
T [K]
302
301
300
299
x [μm]
Energy terms
$\times 10^{13}$
[W/m^3]
1
0
-1
x [μm]
0
0.5
1
1.5
2
2.5
3
Non-isothermal
Isothermal
$J_n \cdot E$
$\frac{3}{2}qn\frac{\theta - V_T}{\tau_E}$
$\frac{dS_n}{dx}$


Va = 0.2 V, τ_E = 0.2 ps,N = 1e+03, μ = const
Electron density
density [m^{-3}]
10^{24}
10^{22}
x [μm]
Electrostatic potential
ψ [V]
0.2
0.1
0
-0.1
x [μm]
SG electron current density
$\times 10^7$
-1.66
-1.67
-1.67
J_n [A/m^2]
x [μm]
Quasi-neutrality ratio
n/N$_D$
1.4
1.2
1
x [μm]
Space charge density
ρ = q(N$_D$-n) [C/m^3]
2000
0
-2000
x [μm]
Temperature
T [K]
302
300
298
x [μm]
Energy terms
$\times 10^{13}$
[W/m^3]
2
0
-2
x [μm]
0
0.5
1
1.5
2
2.5
3
Non-isothermal
Isothermal
$J_n \cdot E$
$\frac{3}{2}qn\frac{\theta - V_T}{\tau_E}$
$\frac{dS_n}{dx}$


Va = 0.3 V, τ_E = 0.2 ps,N = 1e+03, μ = const
Electron density
density [m^{-3}]
10^{24}
10^{23}
10^{22}
0
0.5
1
1.5
2
2.5
3
x [μm]
Electrostatic potential
ψ [V]
0.4
0.2
0
-0.2
x [μm]
SG electron current density
$\times 10^7$
J_n [A/m^2]
-2.49
-2.50
-2.50
-2.51
x [μm]
Quasi-neutrality ratio
n/N$_D$
1.4
1.2
1
x [μm]
Space charge density
ρ = q(N$_D$-n) [C/m^3]
2000
0
-2000
x [μm]
Temperature
T [K]
304
302
300
298
x [μm]
Energy terms
$\times 10^{13}$
[W/m^3]
2
0
-2
x [μm]
Non-isothermal
Isothermal
$J_n \cdot E$
$\frac{3}{2}qn\frac{\theta - V_T}{\tau_E}$
$\frac{dS_n}{dx}$


Va = 0.4 V, τ_E = 0.2 ps,N = 1e+03, μ = const
Electron density
density [m^{-3}]
10^{24}
10^{22}
x [μm]
Electrostatic potential
ψ [V]
0.4
0.2
0
-0.2
x [μm]
SG electron current density
$\times 10^7$
-3.32
-3.33
-3.34
J_n [A/m^2]
x [μm]
Quasi-neutrality ratio
n/N$_D$
1.6
1.4
1.2
1
x [μm]
Space charge density
ρ = q(N$_D$-n) [C/m^3]
2000
0
-2000
x [μm]
Temperature
T [K]
310
305
300
295
x [μm]
Energy terms
$\times 10^{13}$
[W/m^3]
5
0
-5
x [μm]
0
0.5
1
1.5
2
2.5
3
Non-isothermal
Isothermal
$J_n \cdot E$
$\frac{3}{2}qn\frac{\theta - V_T}{\tau_E}$
$\frac{dS_n}{dx}$


Va = 0.5 V, τ_E = 0.2 ps,N = 1e+03, μ = const
Electron density
density [m^{-3}]
10^{24}
10^{22}
x [μm]
Electrostatic potential
ψ [V]
0.4
0.2
0
x [μm]
SG electron current density
$\times 10^7$
-4.15
-4.16
-4.17
J$_n$ [A/m^2]
x [μm]
Quasi-neutrality ratio
n/N$_D$
1.6
1.4
1.2
1
x [μm]
Space charge density
ρ = q(N$_D$-n) [C/m^3]
2000
0
-2000
x [μm]
Temperature
T [K]
310
305
300
x [μm]
Energy terms
$\times 10^{13}$
[W/m^3]
5
0
-5
x [μm]
0
0.5
1
1.5
2
2.5
3
Non-isothermal
Isothermal
$J_n \cdot E$
$\frac{3}{2}qn\frac{\theta - V_T}{\tau_E}$
$\frac{dS_n}{dx}$

6 Tau sweep

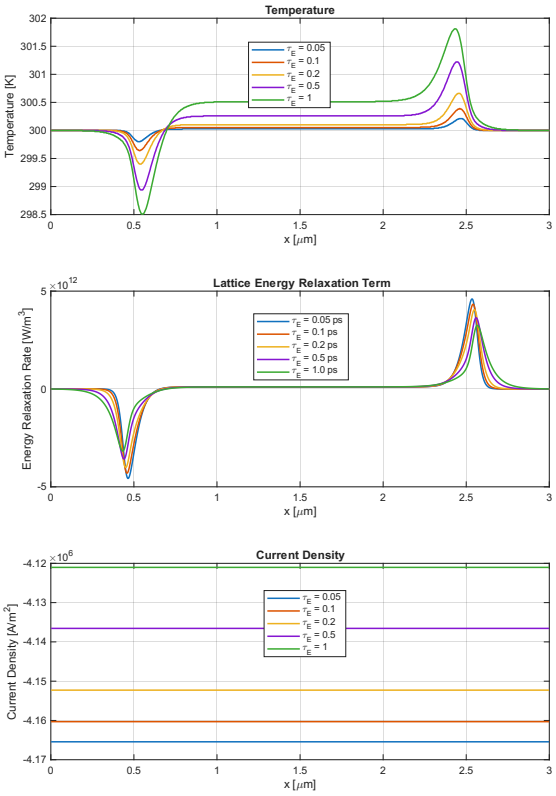


Figure 6: τ sweep for $V_A = 0.05V$

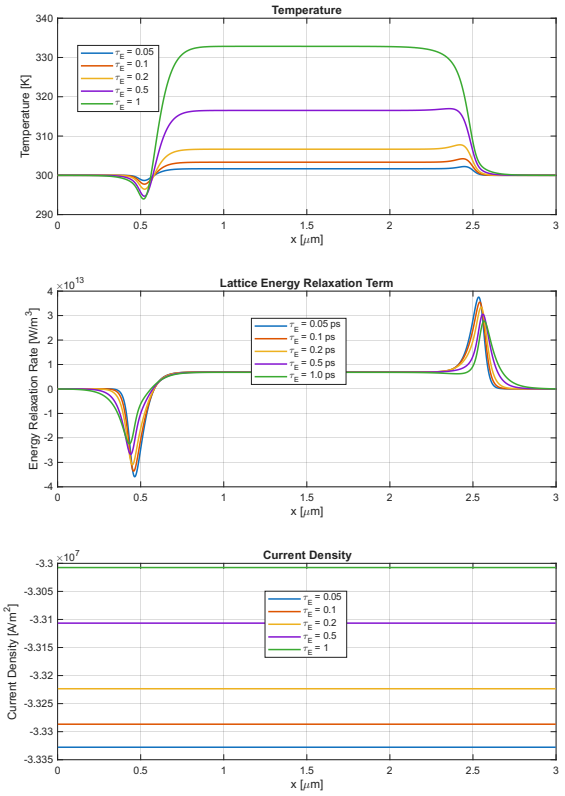


Figure 7: τ sweep for $V_A = 0.4V$

Increasing the energy relaxation time τ_E decouples the mobile electrons from the lattice temperature. Physically, this implies that electrons keep their gained kinetic energy for a longer duration, as the rate at which they dissipate excess energy to the crystal lattice is significantly reduced.
Mathematically, a larger τ_E suppresses the thermal relaxation term within the stationary energy-balance equation:

$$\frac{dS_n}{dx} = J_n E - \frac{3}{2} qn \frac{\vartheta - V_T}{\tau_E}$$

So the localized Joule heating term ($J_n E$) is allowed to dominate the local energy profile. And we have a delayed energy transfer to the lattice, which causes the electron temperature T_n to spike significantly (peaking up to 332.8 K under a 0.4 V bias for $\tau_E = 1$ ps).
Microscopically, the spatial variation of the electron thermal voltage generates an internal thermoelectric force. This force acts as a physical counter-current against the primary electrostatic drift, culminating in a net reduction of the overall current density magnitude $|J_n|$.

	$\tau_E = 0.05$	$\tau_E = 0.1$	$\tau_E = 0.2$	$\tau_E = 0.5$	$\tau_E = 1$
T_n [K]	300.2	300.4	300.7	301.2	301.8
J_n $[A/m^2]$	-4.165 $\cdot 10^6$	-4.160 $\cdot 10^6$	-4.152 $\cdot 10^6$	-4.137 $\cdot 10^6$	-4.121 $\cdot 10^6$
$\frac{3}{2} qn \frac{\vartheta - V_T}{\tau_E}$ $[W/m^3]$	4.599 $\cdot 10^{12}$	4.330 $\cdot 10^{12}$	4.050 $\cdot 10^{12}$	3.635 $\cdot 10^{12}$	3.250 $\cdot 10^{12}$

Table 2: $V_A = 0.05V$

Moreover, we have to consider that highly energized carriers experience severe lattice scattering that degrades their effective mobility.

	$\tau_E = 0.05$	$\tau_E = 0.1$	$\tau_E = 0.2$	$\tau_E = 0.5$	$\tau_E = 1$
T_n [K]	302.2	304.2	307.8	317.0	332.8
J_n $[A/m^2]$	-3.333 $\cdot 10^7$	-3.329 $\cdot 10^7$	-3.322 $\cdot 10^7$	-3.311 $\cdot 10^7$	-3.301 $\cdot 10^7$
$\frac{3}{2}qn\frac{\vartheta - V_T}{\tau_E}$ $[W/m^3]$	3.755 $\cdot 10^{13}$	3.551 $\cdot 10^{13}$	3.344 $\cdot 10^{13}$	3.054 $\cdot 10^{13}$	2.778 $\cdot 10^{13}$

Table 3: $V_A = 0.4$ V

If the mobility is modeled as a function of the local electron temperature, the carrier transport and energy equations become highly coupled. So an increased τ allows the electron temperature to spike near the extracting junction, and this localized thermal peak directly causes a degradation in carrier mobility.

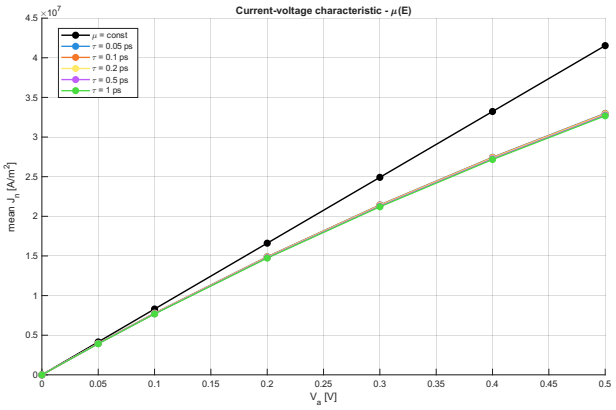


Figure 8: I-V curve with $\mu(E)$

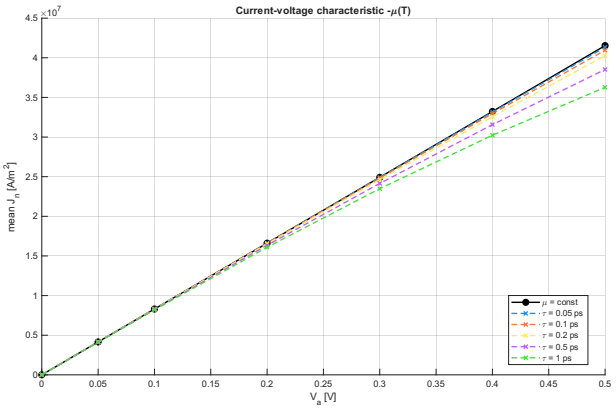


Figure 9: I-V curve with $\mu(T)$

This temperature-induced mobility degradation is clearly demonstrated in Figure 9 : as the relaxation time is increased from 0.05 ps to 1 ps, the mean J_n exhibits a progressive and substantial suppression at higher applied voltages.

On the other hand, if the mobility is modeled only as a function of the local electric field, utilizing classic velocity saturation formulas, the transport equations remain artificially uncoupled from the thermodynamic energy swings. Under this field-dependent assumption, increasing τ still produces the exact same spikes in electron temperature, but the mobility remains completely blind to these thermal variations. This decoupling is plainly verified by Figure 8, where the curves for all relaxation times overlap perfectly into a single line.

7 Nint sweep

	Tn	Jn	Poisson residual / Pscale	Poisson residual / $\max\lvert\rho_{int}\rvert$	Energy residual norm
400	300.619	-4.016 $\cdot 10^6$	1.395510 $\cdot 10^{-10}$	4.515684 $\cdot 10^{-10}$	1.788518 $\cdot 10^{-10}$
800	300.622	-4.012 $\cdot 10^6$	1.419458 $\cdot 10^{-10}$	3.114371 $\cdot 10^{-10}$	1.610664 $\cdot 10^{-10}$
1600	300.622	-4.009 $\cdot 10^6$	1.425084 $\cdot 10^{-10}$	2.650704 $\cdot 10^{-10}$	2.519135 $\cdot 10^{-10}$

Table 4: $V_A = 0.05V$

	Tn	Jn	Poisson residual / Pscale	Poisson residual / $\max\lvert\rho_{int}\rvert$	Energy residual norm
400	304.859	-2.410 $\cdot 10^7$	9.021018 $\cdot 10^{-10}$	2.907524 $\cdot 10^{-9}$	1.040326 $\cdot 10^{-9}$
800	304.863	-2.407 $\cdot 10^7$	9.196862 $\cdot 10^{-10}$	2.012180 $\cdot 10^{-9}$	9.410900 $\cdot 10^{-10}$
1600	304.862	-2.406 $\cdot 10^7$	9.245786 $\cdot 10^{-10}$	1.715874 $\cdot 10^{-9}$	8.662792 $\cdot 10^{-10}$

Table 5: $V_A = 0.3V$

The spatial convergence analysis confirms that the numerical method is highly stable, accurate, and grid-independent. By systematically refining the mesh from 400 to 800 and then 1600 nodes, macroscopic quantities

such as the current density and the peak T_n rapidly stabilize towards their asymptotic values. For example, under a 0.3 V bias, the variations in J_n and T_n between 800 and 1600 nodes are virtually negligible compared to the initial refinement step, demonstrating that the steep gradients at the device junctions are properly resolved, and local truncation errors are minimized. The robustness of the scheme is further validated by the normalized Poisson and Energy residuals, which are strictly maintained between 10^{-10} and 10^{-9}. Furthermore, as shown by the convergence rates, the method consistently achieves the expected robust second-order spatial accuracy ($p \approx 2.12$) for all primary variables across the entire active bias sweep.

V_a [V]	p_n	p_φ	p_θ
0	2.12400	2.12569	0.00244
0.05	2.12399	2.12577	2.12724
0.1	2.12398	2.12582	2.12801
0.2	2.12396	2.12559	2.12780
0.3	2.12392	2.12560	2.12591
0.4	2.12388	2.12706	2.12341
0.5	2.12383	2.12839	2.12549

Table 6: Convergence rates at varying bias voltages.

A notable anomaly occurs at thermal equilibrium, where the calculated convergence rate for the temperature (p_θ) drops to 0.00244. This is not a failure of the numerical scheme, but a consequence of the physics: at zero bias, $J_n = 0$ and the electron temperature is constrained to perfectly match the lattice temperature everywhere. Because the continuous energy balance is satisfied and the true analytical error in temperature is effectively at the limit of machine precision, the mathematical computation becomes dominated by numerical noise.

8 Small device analysis

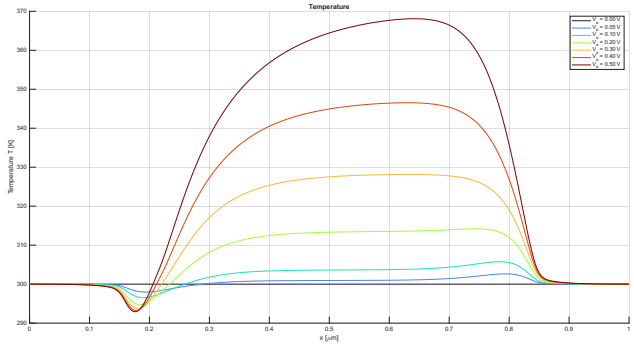


Figure 10: Temperature

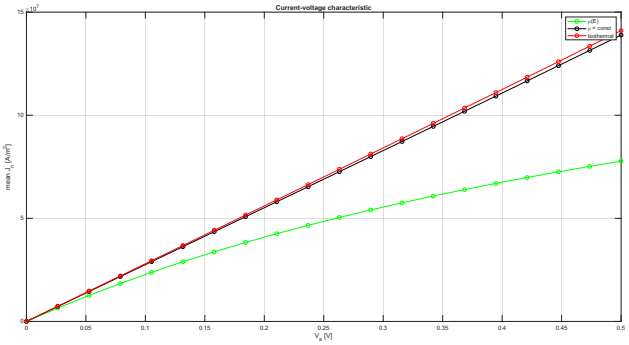


Figure 11: I-V curve

When transitioning to a shorter device ($L = 1\,\mu$m), the reduced channel length escalates the internal electric field for a given applied voltage V_a, resulting in strong deviations from the baseline behavior. According to the model's validity indicators, a velocity diagnostic ratio $\max(|u|)/v_{\text{sat}}$ above 0.3 make the constant-mobility assumption highly questionable. As observed in the constant-mobility simulation (Table 7), this validity threshold is violated at an early stage (for $V_a = 0.200\,$V, the velocity ratio already reaches 0.3438)

V_a [V]	$\dfrac{\max(\lvert u\rvert)}{v_{\text{sat}}}$	$\dfrac{\max(\mu\lvert\nabla\varphi_F\rvert)}{v_{\text{sat}}}$	$\max(T_n)$	$\dfrac{\text{spread}(J_n)}{J_{\text{scale}}}$	$\dfrac{\text{spread}(J_n)}{\lvert J_{\text{mean}}\rvert}$
0.000	$1.971 \cdot 10^{-13}$	$2.080 \cdot 10^{-13}$	300.0	$6.168 \cdot 10^{-12}$	$2.869 \cdot 10^{1}$
0.050	$8.583 \cdot 10^{-2}$	$8.586 \cdot 10^{-2}$	302.5	$5.739 \cdot 10^{-9}$	$1.727 \cdot 10^{-7}$
0.100	$1.717 \cdot 10^{-1}$	$1.718 \cdot 10^{-1}$	305.9	$2.642 \cdot 10^{-9}$	$3.974 \cdot 10^{-8}$
0.200	$3.438 \cdot 10^{-1}$	$3.439 \cdot 10^{-1}$	318.3	$1.527 \cdot 10^{-8}$	$1.147 \cdot 10^{-7}$
0.300	$5.168 \cdot 10^{-1}$	$5.166 \cdot 10^{-1}$	341.2	$3.249 \cdot 10^{-8}$	$1.624 \cdot 10^{-7}$
0.400	$6.920 \cdot 10^{-1}$	$6.900 \cdot 10^{-1}$	373.3	$7.752 \cdot 10^{-8}$	$2.899 \cdot 10^{-7}$
0.500	$8.713 \cdot 10^{-1}$	$8.641 \cdot 10^{-1}$	414.6	$1.079 \cdot 10^{-7}$	$3.218 \cdot 10^{-7}$

Table 7: Small device with constant mobility

To correct these high-field transport anomalies, a velocity-saturation correction is introduced to implement a field-dependent mobility description. With this adjustment, the framework transitions toward a robust high-field drift-diffusion description, meaning the model remains structurally valid for any processed value of the velocity diagnostic ratio $\max(|u|)/v_{\mathrm{sat}}$, provided it does not exceed the absolute physical limit of 1. The impact of this correction is clearly demonstrated in the field-dependent simulation parameters (Table 8). By regularizing carrier velocities under big local fields, the overestimation of energy flux is mitigated, which consequently tempers excessive electron heating and clamps the maximum device temperature at 396.4 K.

V_a [V]	$\frac{\max(\lvert u\rvert)}{v_{\mathrm{sat}}}$	$\frac{\max(\mu\lvert\nabla\varphi_F\rvert)}{v_{\mathrm{sat}}}$	$\max(T_n)$	$\frac{\mathrm{spread}(J_n)}{J_{\mathrm{scale}}}$	$\frac{\mathrm{spread}(J_n)}{\lvert J_{\mathrm{mean}}\rvert}$
0.000	$1.971 \cdot 10^{-13}$	$2.080 \cdot 10^{-13}$	300.0	$6.168 \cdot 10^{-12}$	$2.869 \cdot 10^{1}$
0.050	$8.556 \cdot 10^{-2}$	$8.559 \cdot 10^{-2}$	302.4	$5.270 \cdot 10^{-9}$	$1.591 \cdot 10^{-7}$
0.100	$1.696 \cdot 10^{-1}$	$1.697 \cdot 10^{-1}$	305.9	$1.169 \cdot 10^{-8}$	$1.781 \cdot 10^{-7}$
0.200	$3.282 \cdot 10^{-1}$	$3.282 \cdot 10^{-1}$	317.7	$2.398 \cdot 10^{-8}$	$1.887 \cdot 10^{-7}$
0.300	$4.692 \cdot 10^{-1}$	$4.688 \cdot 10^{-1}$	338.2	$4.102 \cdot 10^{-8}$	$2.259 \cdot 10^{-7}$
0.400	$5.937 \cdot 10^{-1}$	$5.899 \cdot 10^{-1}$	364.8	$6.137 \cdot 10^{-8}$	$2.687 \cdot 10^{-7}$
0.500	$7.074 \cdot 10^{-1}$	$6.941 \cdot 10^{-1}$	396.4	$2.853 \cdot 10^{-8}$	$1.063 \cdot 10^{-7}$

Table 8: Small device with field-dependent mobility

9 Conclusion

The non-isothermal unipolar energy-balance simulation characterizes the coupled electrothermal dynamics of the $n^+ - n - n^+$ semiconductor device. Under thermal equilibrium conditions, the framework achieves high spatial consistency, capturing the exact balance between drift and diffusion mechanisms and the carrier spillover across the graded doping transitions. Under active bias, the device exhibits a highly asymmetric thermal profile consisting of a thermoelectric cooling dip near the injecting junction and a temperature peak at the extracting interface. Grid refinement sweeps confirm the numerical stability and second-order spatial accuracy of the decoupled Gummel iterative scheme under active transport. In addition, short-channel analysis at $L = 1\,\mu$m highlights the breakdown of the constant low-field mobility model under high driving fields, which is effectively regularized by field-dependent velocity-saturation corrections. Ultimately, the study demonstrates that simple phenomenological mobility descriptions cannot be combined without double-counting scattering events, underscoring the need for full hydrodynamic equations or direct solutions to the Boltzmann Transport Equation in deep submicron regimes experiencing severe carrier heating.